

Milestone 2 Mini-capstone: Project Planning and System Design (15% (continuation from Milestone #1))

Group: Same as the previous milestone

Objective

In this milestone, your group will select a project topic and create the planning and design documents that will guide the development of your application in Milestone 2.

The milestone is worth **15% of the final grade**. The **submission deadline is listed in the Instructional Plan**, and each group will deliver a **short in-class presentation**. All group members must participate.

Project Requirements

1. Select a Project Topic (same as milestone #1)

Choose a project that can be developed as a full-stack web application (e.g., Library System, Event Management, Online Store, Fitness Tracker, Course Registration System, etc.).

2. Technology

Database: MongoDB, MySQL, **or** PostgreSQL

Front-end: At this stage, your front-end must be based on HTML/CSS/JavaScript (no API's just yet)

Back end: TypeScript + Express backend

Explain why your team selected the database technology.

3. Design the Database (ER diagram, table description)

Plan a database with **collections (MongoDB)** or at least **3 tables (MySQL/PostgreSQL)**.

For each table/collection, include:

- Name
- Purpose
- Fields
- Data Types
- Relationships

Create an ERD (Entity-Relationship Diagram) or a database diagram.

Recommended Tools (hands-on design):

- MySQL Workbench
- MongoDB Compass Schema Design
- draw.io (diagrams.net)
- dbdiagram.io

Note: AI-generated diagrams are not acceptable. Students should be able to explain every table, field, and relationship during the presentation.

4. Design the Frontend (Wireframes)

Plan the pages your application will contain.

Minimum:

- Home Page
- Data Entry Page
- Data Listing Page
- Edit/Update Page

Create wireframes for each page, showing its layout.

Recommended Tools (choose one):

- Figma, draw.io, Canva, or Pencil Project

Students should be prepared to explain their design choices.

5. Plan the Backend API (routes description and explanation for each page)

Identify the main routes your application will need.

Example:

- GET /products
- POST /products
- PUT /products/:id
- DELETE /products/:id

Briefly describe what each route will do.

6. GitHub Repository

Use the same GitHub repository created during Milestone #1.

The repository must contain:

- Project description
- Database design
- Wireframes
- API plan/routes
- README.md

Each student must contribute commits.

Team Responsibilities

Work must be divided among all three group members.

Student 1: Database Designer

Responsible for:

- Database selection
- Table/collection design
- ERD creation
- Database documentation

Student 2: Backend Planner

Responsible for:

- API route planning
- Express architecture planning
- Technical documentation

Student 3: Frontend Designer

Responsible for:

- Wireframes
- User interface planning
- Navigation design

All members are expected to participate in project discussions and the presentation.

Presentation (15 Minutes)

Each group will present:

1. Project idea
2. Database design
3. Frontend wireframes
4. Planned API routes
5. Team responsibilities

Every student must explain their assigned section.

Students should be prepared to answer questions about their own work.

Deliverables

Submit:

- GitHub Repository Link (add your instructor)
 - PDF Report containing:
 - Project Description
 - Database Design
 - ERD/Database Diagram
 - Frontend Wireframes
 - Planned API routes
 - Team Responsibilities
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Important: looking ahead

The designs created in this milestone will be used in Milestone 3.